

Fraud & Anomaly Detection

Cause-first risk framework using real G'Contest banking data

1,418,030

Transactions scored

402,266

High/Critical cases

38,756

Customers impacted

- No synthetic data and no fake fraud labels.
- Output is a prevention queue with explainable risk reasons.
- Framework aligns with risk-based banking controls: step-up authentication, manual review, AML escalation.
- Optimization priority: catch more fraud-risk cases first, while reporting false-positive rate for customer experience control.

Why Cause-First

BGK note: identify root causes before introducing the model

- Account takeover / identity compromise: new device, new IP, night access, new beneficiary, late-stage digital activity.
- Unauthorized transfer / capital outflow: outside-bank transfer, amount above customer baseline, daily burst, cash-out versus CASA.
- AML network / mule-account pattern: shared IP/device/beneficiary, round high-value transfers, repeated external transfers.

Framework

Data dictionary -> Customer 360 baseline -> root-cause rules -> weak labels -> prevention model

1. Data Customer, Transaction, Activity, Deposit, Lending, Card

2. Customer 360 Transactional, Financial, Environmental, Behavioral baseline

3. Thresholds Rolling 30/60/90 days plus personalized $IQR = Q3 + 1.5 * IQR$

4. Model Rules create weak labels; supervised model learns prevention risk

5. Hybrid Action Rule+ML matrix -> Allow / Watchlist / Step-up / Block

Prevention Performance

Weak-label evaluation for an operational fraud prevention queue

100.0%

Recall / prevention coverage

100.0%

Precision of Step-up/Block

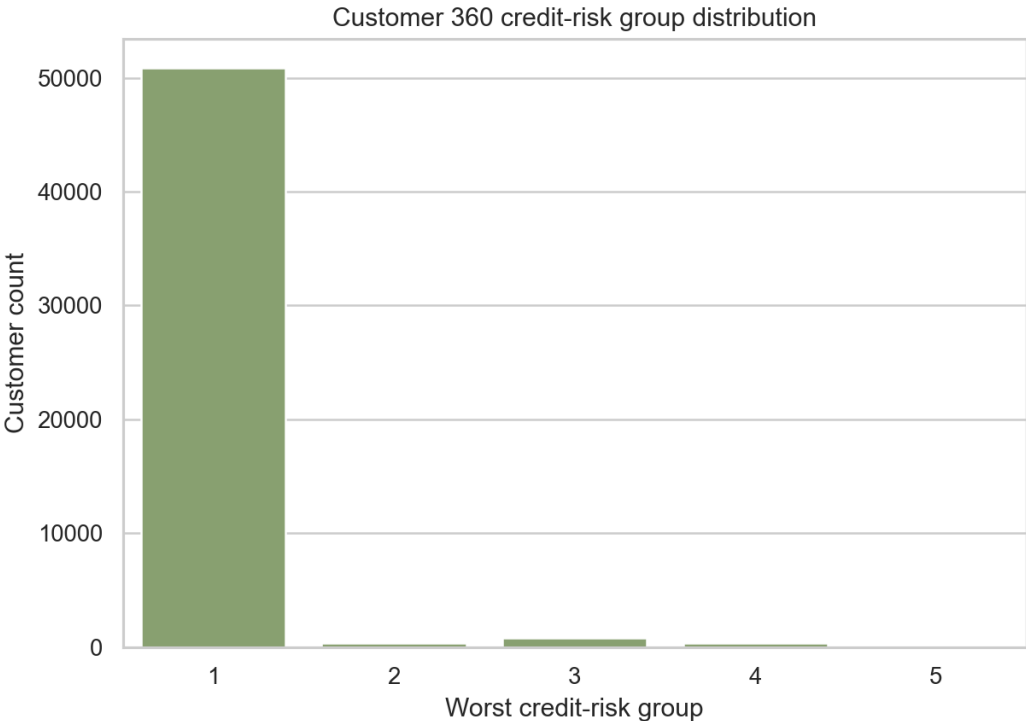
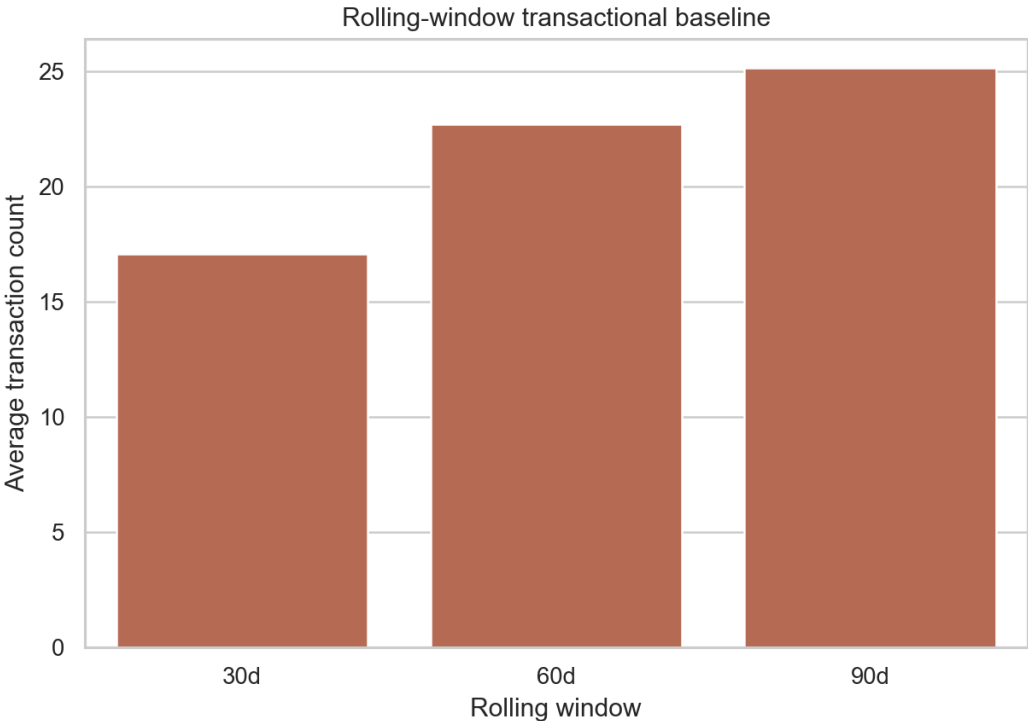
0.00%

False-positive rate

- Metrics are measured against rule-derived weak labels because the contest data has no confirmed fraud labels.
- The bank objective is risk minimization: prioritize catching suspected fraud, then tune thresholds against review capacity and customer friction.
- Protected amount by Block/Step-up queue: 6,935,381,708,289 VND.
- High-threshold confusion matrix: {'true_negative': 43333, 'false_positive': 0, 'false_negative': 2167, 'true_positive': 19500}.

Customer 360 Baseline

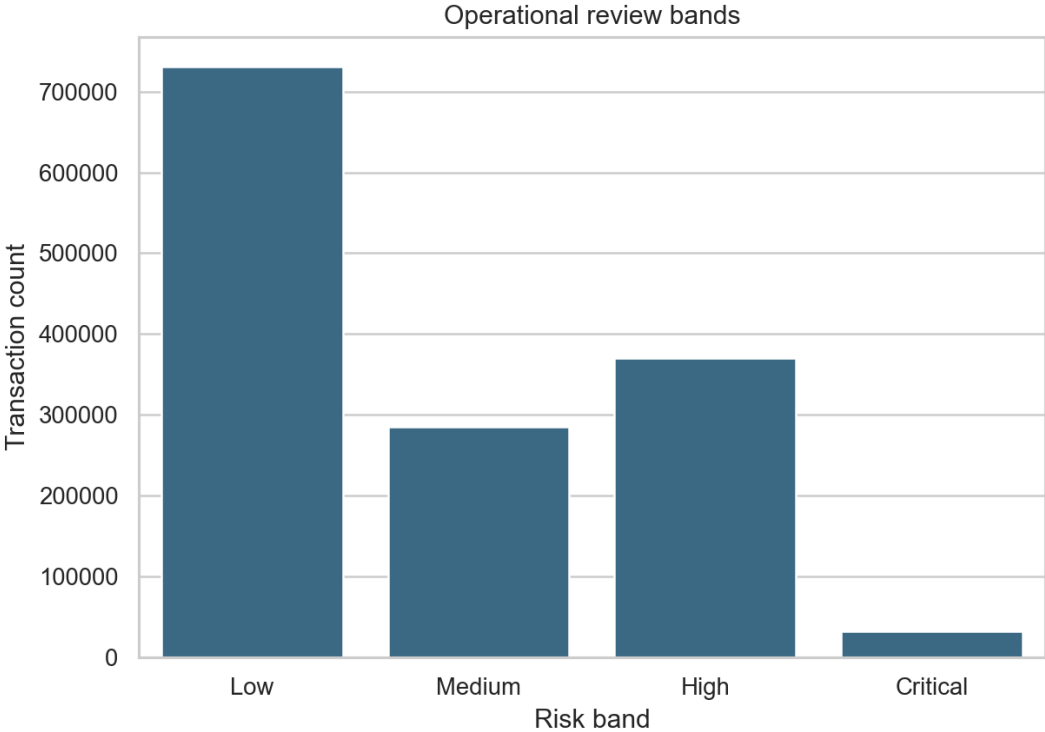
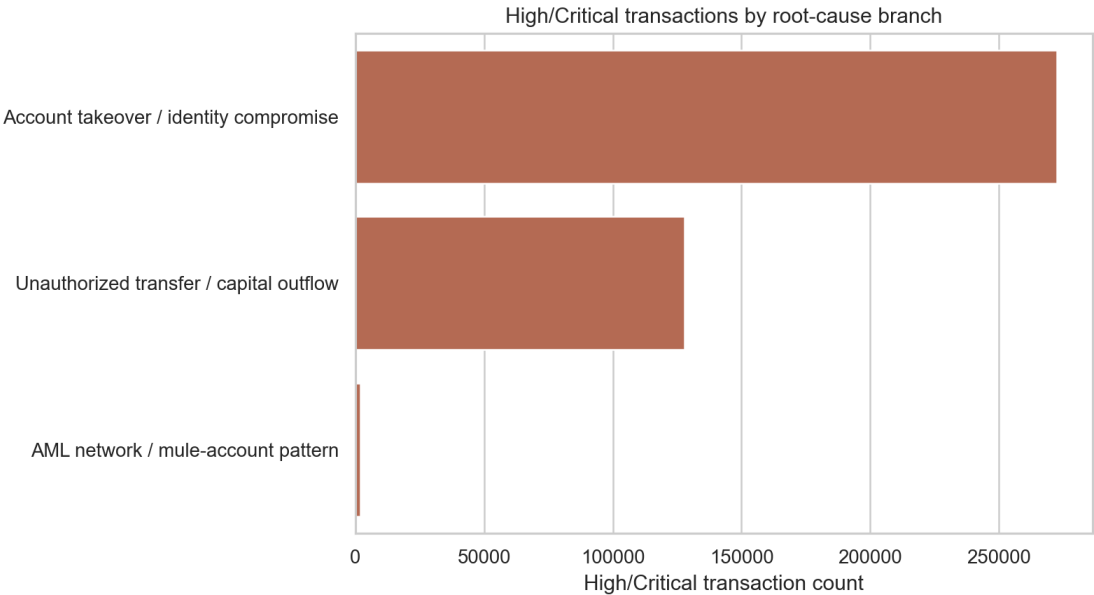
One customer, one current financial-behavioral profile



- Transactional: amount average/P95/IQR, golden hour, rolling 30/60/90-day activity.
- Financial: CASA/TD balance, card utilization, overdue and credit-risk group.
- Environmental and behavioral: trusted device/IP, known beneficiary, app activity, night/late-stage activity

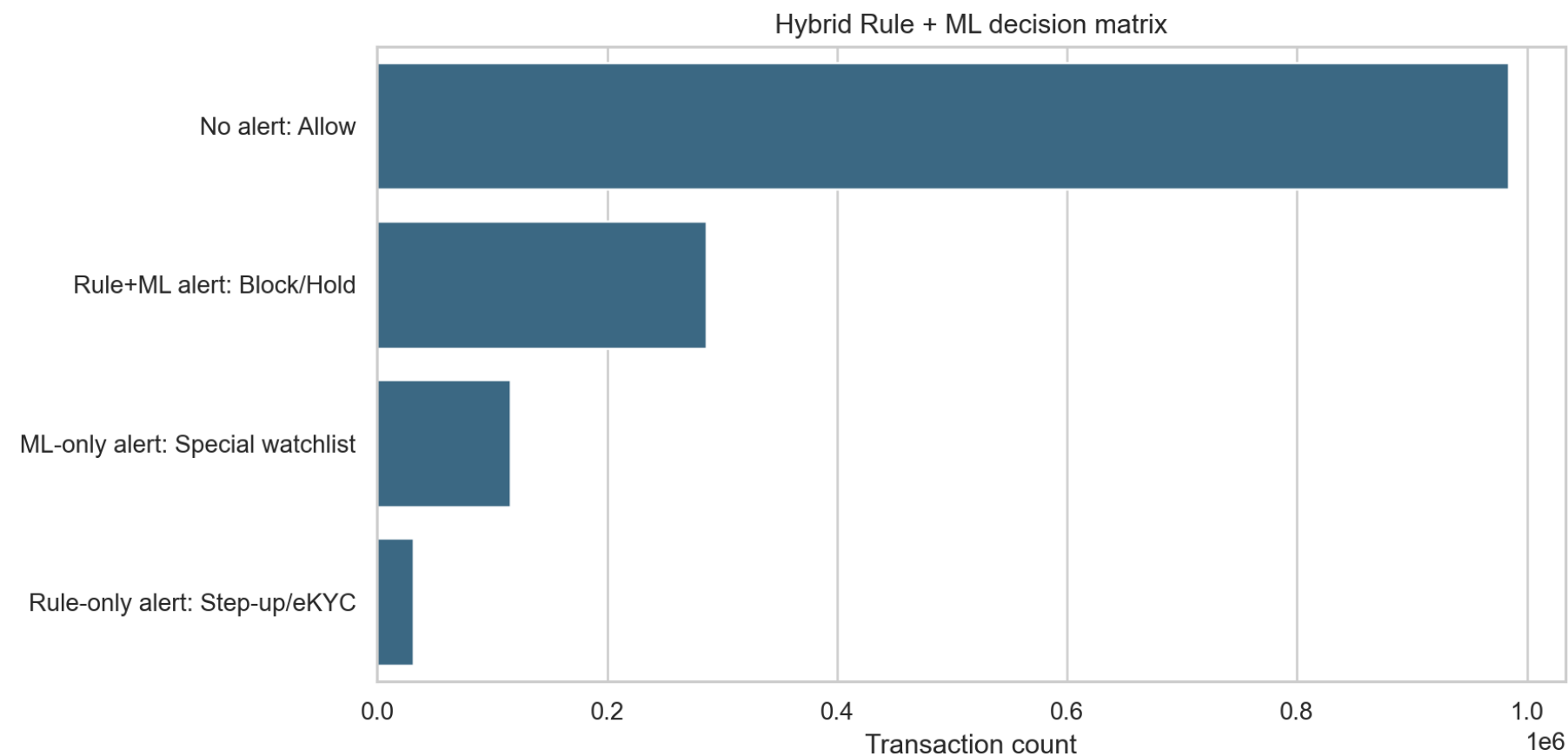
Key Insights

Top-risk queue is explainable by root-cause branch



Hybrid Decision Matrix

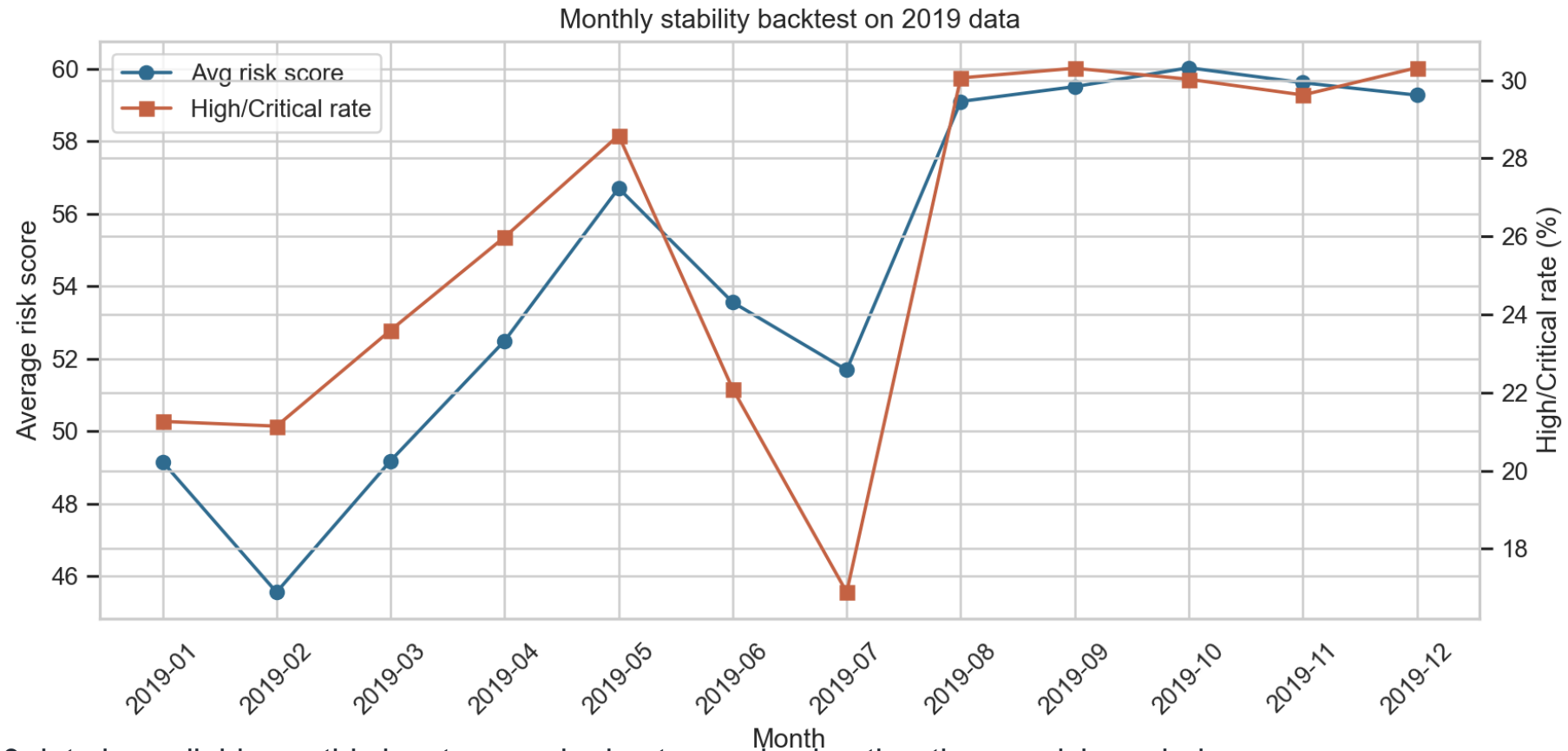
Rule engine and ML confidence are converted into bank actions



- Rule+ML alert: Block/Hold.
- Rule-only alert: Step-up/eKYC to reduce false positives.
- ML-only alert: Special watchlist for patterns not yet covered by rules.
- No alert: Allow while baseline continues to update.

Temporal Stability

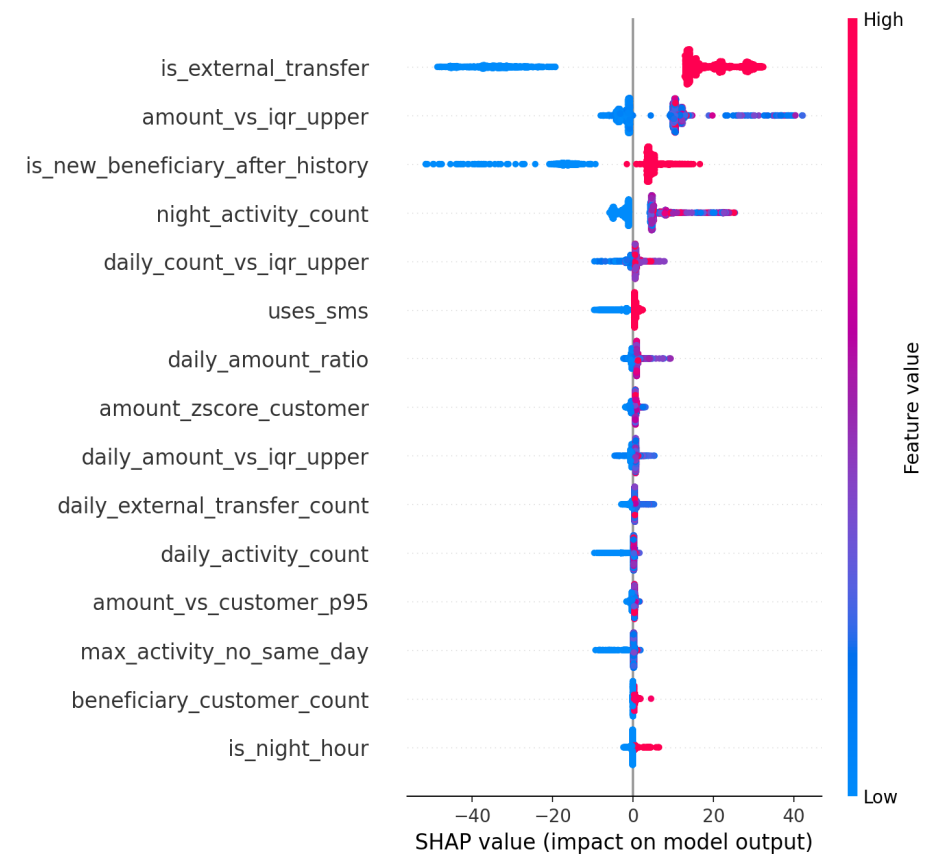
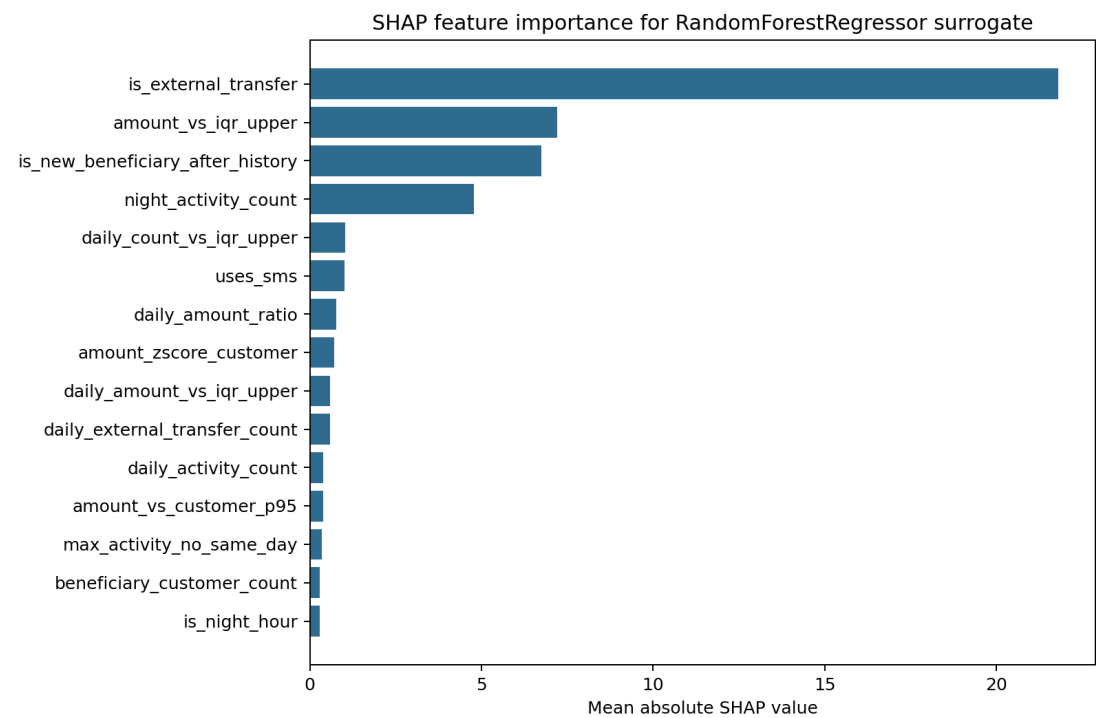
2019 monthly/quarterly backtest for review-rate stability



- Only 2019 data is available, so this is a temporal robustness check rather than a crisis-period validation.
- The KPI is not confirmed-fraud accuracy; it is weak-label prevention coverage, protected amount, and reason-code consistency over time.

xAI Engine

SHAP explains a tree surrogate of the supervised prevention score



Operational Recommendations

Turn model output into bank actions

- Critical: temporary hold, near-real-time manual review, customer verification, step-up authentication.
- High: step-up authentication before allowing the transaction to proceed.
- AML branch: network escalation rather than single-transaction review only.
- Medium: enhanced monitoring; escalate if repeated within the next review window.
- Feedback loop: investigator decisions become labels for Precision@K, Recall@K and supervised model tuning.

Live Demo

Streamlit advisor for transaction/customer review

- Run ``streamlit run src/demo_app.py``.
- Select a top-risk transaction or enter a CUSTOMER_NUMBER.
- The advisor returns risk band, root-cause branch, reason codes, recommended action and SHAP evidence.
- This is designed for a short judging demo or screen-recorded walkthrough.